

Qualification Pack



Technician 5G - Active Network Installation

QP Code: TEL/Q6213

Version: 3.0

NSQF Level: 4

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TEL/Q6213: Technician 5G - Active Network Installation

Brief Job Description

A 5G Technician -Active Network Installation is responsible for installing 5G network equipment at the rack level and performing active network installations. The role also involves identifying and rectifying faults or malfunctions that may arise during the installation process.

Personal Attributes

The individual should possess strong attention to detail, analytical thinking, and problem-solving skills. Effective coordination with team members is essential, along with good verbal and written communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N6104: Carry out Rack Level Installation](#)
2. [TEL/N6105: Carry out 5G active network installation](#)
3. [TEL/N6246: Follow the Occupational Health and Safety Instructions during Tower Climbing](#)
4. [TEL/N9105: Follow sustainable practices in telecom infrastructure installation](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network (Active Components Installation)
Country	India
NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.0802

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Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Electronics Engineering / Electrical Engineering/ Mobile and Wireless Communication/ Networking and Telecom Systems) OR 10th grade pass with 3 Years of experience in the telecom infrastructure domain OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience in the telecom infrastructure domain OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in the telecom infrastructure domain
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-TL-04089-2025-V2-TSSC
NQR Version	2

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TEL/N6104: Carry out Rack Level Installation

Description

This unit is about performing rack-level installation for 5G network equipment while ensuring industry-standard safety, power, and cooling requirements.

Scope

The scope covers the following :

- Prepare for the installation of racks and equipment for 5G networks
- Install and secure racks and equipment for 5G networks
- Conduct post-installation verification and troubleshooting

Elements and Performance Criteria

Prepare for the installation of racks and equipment for 5G networks

To be competent, the user/individual on the job must be able to:

- PC1.** assess the designated installation area to verify space availability, load-bearing capacity, and future expansion feasibility
- PC2.** check facility entry points, door heights, and floor strength to facilitate safe movement and installation of racks and equipment
- PC3.** identify and select appropriate rack types (open frame, enclosed, wall-mount) based on the equipment to be installed and ease of cable and device management
- PC4.** ensure power supply circuits provide the required voltage, amperage, and redundancy for uninterrupted operation of 5G network equipment
- PC5.** verify the absence of heat sources in and around the installation area to prevent thermal risks
- PC6.** assess the adequacy of active or passive cooling systems, including ducts, dropped ceilings, or Computer Room Air Conditioners (CRACs), to manage heat dissipation
- PC7.** develop a layout plan for rack placement, considering power distribution, ventilation pathways, and future equipment scalability
- PC8.** confirm the availability of proper grounding and earthing systems to prevent electrical hazards

Install and secure racks and equipment for 5G networks

To be competent, the user/individual on the job must be able to:

- PC9.** arrange racks in a hot-aisle/cold-aisle configuration to enhance cooling efficiency and reduce energy consumption
- PC10.** position and secure equipment inside racks, ensuring optimal space utilization, accessibility for maintenance, and future expansion
- PC11.** inspect received gNodeB and other 5G network components for physical damage or defects before installation
- PC12.** report and coordinate with vendors or support teams for the replacement or repair of damaged/faulty gNodeB and other components

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- PC13.** mount and secure gNodeB and associated 5G equipment using appropriate fasteners while adhering to rack load ratings
- PC14.** position heavier equipment at the lower sections of racks to maintain stability and prevent top-heavy tipping hazards
- PC15.** distribute high-density and high-wattage equipment across multiple racks to minimize thermal hotspots
- PC16.** follow manufacturer guidelines and safety protocols for assembling and installing blade servers and power distribution units (PDUs)
- PC17.** connect power, network, and grounding cables following structured cabling best practices to ensure safety and organization

Conduct post-installation verification and troubleshooting

To be competent, the user/individual on the job must be able to:

- PC18.** perform post-installation inspection to confirm proper mounting, cabling, and power connectivity of installed equipment
- PC19.** identify and rectify common installation errors, such as improper cable routing, loose connections, or incorrect equipment placement
- PC20.** run initial power-on and basic operational tests for installed 5G network equipment
- PC21.** troubleshoot hardware and connectivity issues following manufacturer guidelines and escalate unresolved problems to relevant teams
- PC22.** configure operating system and virtual machine (VM) settings as per standard deployment procedures
- PC23.** document installation details, including rack layout, power consumption, and network configurations, for future reference

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types, dimensions, and load ratings of racks used for IT and 5G network equipment installation
- KU2.** the standard dimensions and specifications of 5G network components, including gNodeB, switches, routers, and servers
- KU3.** key rack components such as mounting rails, doors, ventilation panels, and cable management accessories
- KU4.** the criteria for selecting an optimal installation space, including environmental, structural, and electrical considerations
- KU5.** the methods for evaluating power circuit capabilities, including voltage, amperage, redundancy, and surge protection
- KU6.** the importance of effective cooling solutions, including hot-aisle/cold-aisle design, airflow management, and HVAC integration
- KU7.** procedures for arranging and securing network equipment for stability, maintenance, and expandability
- KU8.** common causes of thermal issues and techniques to prevent overheating of 5G network equipment

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- KU9.** best practices for power and network cable management to ensure safety and ease of maintenance
- KU10.** techniques for inspecting and identifying faults in gNodeB and other 5G hardware components
- KU11.** installation and safety procedures for high-power and high-density server equipment
- KU12.** industry best practices for structured cabling, grounding, and earthing in telecommunications environments
- KU13.** basic principles of networking, including IP addressing, TCP/IP, VLANs, and remote access protocols
- KU14.** common troubleshooting techniques for hardware faults, connectivity failures, and power issues
- KU15.** steps to configure virtualization environments (e.g., VMware) for network deployment and testing
- KU16.** documentation standards for recording installation layouts, cable routing, and equipment configurations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain clear and accurate documentation of installation, configuration, and troubleshooting activities
- GS2.** read and interpret technical manuals, schematics, and vendor guidelines related to 5G network installation
- GS3.** communicate professionally and effectively with team members, vendors, and technical support personnel
- GS4.** plan and prioritize tasks to ensure efficient completion of installation activities within deadlines
- GS5.** work collaboratively with coworkers to ensure seamless coordination during large-scale equipment installations
- GS6.** apply critical thinking to evaluate installation challenges and implement the most effective solutions
- GS7.** follow safety protocols and emergency procedures to prevent workplace accidents during installation
- GS8.** adapt to new technologies and industry standards by keeping up to date with evolving network hardware trends
- GS9.** use digital tools and software applications for installation planning, troubleshooting, and configuration
- GS10.** analyze system performance metrics to identify inefficiencies and recommend improvements post-installation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for the installation of racks and equipment for 5G networks</i>	10	20	-	7
PC1. assess the designated installation area to verify space availability, load-bearing capacity, and future expansion feasibility	1	3	-	1
PC2. check facility entry points, door heights, and floor strength to facilitate safe movement and installation of racks and equipment	2	3	-	1
PC3. identify and select appropriate rack types (open frame, enclosed, wall-mount) based on the equipment to be installed and ease of cable and device management	2	3	-	1
PC4. ensure power supply circuits provide the required voltage, amperage, and redundancy for uninterrupted operation of 5G network equipment	1	3	-	0.5
PC5. verify the absence of heat sources in and around the installation area to prevent thermal risks	1	2	-	0.5
PC6. assess the adequacy of active or passive cooling systems, including ducts, dropped ceilings, or Computer Room Air Conditioners (CRACs), to manage heat dissipation	1	2	-	1
PC7. develop a layout plan for rack placement, considering power distribution, ventilation pathways, and future equipment scalability	1	2	-	1
PC8. confirm the availability of proper grounding and earthing systems to prevent electrical hazards	1	2	-	1
<i>Install and secure racks and equipment for 5G networks</i>	12	20	-	8
PC9. arrange racks in a hot-aisle/cold-aisle configuration to enhance cooling efficiency and reduce energy consumption	2	3	-	1
PC10. position and secure equipment inside racks, ensuring optimal space utilization, accessibility for maintenance, and future expansion	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. inspect received gNodeB and other 5G network components for physical damage or defects before installation	2	2	-	1
PC12. report and coordinate with vendors or support teams for the replacement or repair of damaged/faulty gNodeB and other components	1	2	-	1
PC13. mount and secure gNodeB and associated 5G equipment using appropriate fasteners while adhering to rack load ratings	1	2	-	1
PC14. position heavier equipment at the lower sections of racks to maintain stability and prevent top-heavy tipping hazards	1	2	-	1
PC15. distribute high-density and high-wattage equipment across multiple racks to minimize thermal hotspots	1	2	-	1
PC16. follow manufacturer guidelines and safety protocols for assembling and installing blade servers and power distribution units (PDUs)	1	2	-	0.5
PC17. connect power, network, and grounding cables following structured cabling best practices to ensure safety and organization	1	2	-	0.5
<i>Conduct post-installation verification and troubleshooting</i>	8	10	-	5
PC18. perform post-installation inspection to confirm proper mounting, cabling, and power connectivity of installed equipment	2	2	-	1
PC19. identify and rectify common installation errors, such as improper cable routing, loose connections, or incorrect equipment placement	2	2	-	1
PC20. run initial power-on and basic operational tests for installed 5G network equipment	1	2	-	1
PC21. troubleshoot hardware and connectivity issues following manufacturer guidelines and escalate unresolved problems to relevant teams	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. configure operating system and virtual machine (VM) settings as per standard deployment procedures	1	1	-	0.5
PC23. document installation details, including rack layout, power consumption, and network configurations, for future reference	1	1	-	0.5
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6104
NOS Name	Carry out Rack Level Installation
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network (Active Components Installation)
NSQF Level	4
Credits	5
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N6105: Carry out 5G active network installation

Description

This unit is about performing various types of cabling, installations, and commissioning activities for 5G active network infrastructure.

Scope

The scope covers the following :

- Carry out power, earthing, and RF cabling
- Install, configure, and commission backhaul connectivity

Elements and Performance Criteria

Carry out power, earthing and RF cabling

To be competent, the user/individual on the job must be able to:

- PC1.** ensure the availability of required installation materials as per the installation document, including terminal connectors, thimbles, surge protectors, and cable ties
- PC2.** assess the power requirements of the gNodeB and associated equipment, ensuring compatibility with the available power supply
- PC3.** select appropriate power, earthing, and RF cables based on specifications, load requirements, and environmental conditions
- PC4.** inspect all cables for any physical damage or manufacturing defects and coordinate replacements, if necessary
- PC5.** install and secure power cables from the power source to the equipment, ensuring compliance with safety and regulatory standards
- PC6.** establish proper earthing connections, measuring and maintaining earth resistance values within permissible limits
- PC7.** install RF cables between the gNodeB, other radio equipment, and antennas, ensuring minimal signal loss
- PC8.** route and terminate cables correctly to omnidirectional and sector antennas while allowing flexibility for maintenance and upgrades
- PC9.** conduct continuity and performance tests on installed cables using industry-standard tools and rectify issues if detected
- PC10.** implement shielding and grounding techniques to mitigate electromagnetic interference (EMI) and ensure optimal signal transmission

Install, configure, and commission backhaul connectivity

To be competent, the user/individual on the job must be able to:

- PC11.** coordinate with relevant personnel to establish high-throughput Ethernet or fiber-optic backhaul connectivity
- PC12.** deploy fiber optic solutions for backhaul connectivity, ensuring proper termination and splicing techniques

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- PC13.** validate and configure backhaul connectivity, ensuring seamless communication between gNodeB and the central office
- PC14.** implement Centralised Unit-Distributed Unit (CU-DU) split architecture for optimized network functionality
- PC15.** configure gNodeB settings to enable integration with the Centralised Unit (CU) and core network
- PC16.** ensure gNodeB visibility in the central unit for remote commissioning, monitoring, and control
- PC17.** install and integrate environmental alarm systems with the central monitoring unit to enable proactive site management
- PC18.** deploy and configure the required Operating System (OS) and Virtual Machine (VM) environments for network operations
- PC19.** verify network synchronization and timing configurations to align with industry standards and optimize performance
- PC20.** conduct functional tests, including throughput and latency checks, to validate end-to-end connectivity and network performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** industry standards, such as 3rd Generation Partnership Project (3GPP) releases relevant to 5G network deployment
- KU2.** components of 5G gNodeB, their functions, and interoperability requirements
- KU3.** power and grounding principles, including best practices for electrical safety and surge protection
- KU4.** types, specifications, and applications of power, earthing, and RF cables used in 5G installations
- KU5.** procedures for installing and securing power, earthing, and RF cables in compliance with industry regulations
- KU6.** best practices for routing and terminating cables between various types of antennas to minimize losses and interference
- KU7.** tools and techniques for conducting cable transmission tests and identifying common faults
- KU8.** methods for establishing high-capacity Ethernet and fiber-based backhaul connectivity
- KU9.** backhaul network architecture, including CU-DU split, and its impact on network performance
- KU10.** configuration and commissioning processes for gNodeB integration with the central unit
- KU11.** network synchronization techniques, including Precision Time Protocol (PTP) and Global Navigation Satellite System (GNSS) synchronization
- KU12.** security considerations for network elements, including data encryption, firewall settings, and access control
- KU13.** methods for configuring and integrating environmental alarm systems for real-time network monitoring
- KU14.** installation and optimization of OS and VM environments for supporting 5G network functions

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KU15. testing methodologies to evaluate network performance, including latency, packet loss, and signal quality

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** document work activities, configurations, and testing results accurately for future reference and compliance
- GS2.** read and interpret technical manuals, installation guides, and schematic diagrams
- GS3.** communicate effectively with team members, supervisors, and external stakeholders
- GS4.** listen actively and follow instructions precisely to ensure accurate installation and commissioning
- GS5.** organize and prioritize tasks efficiently to meet project deadlines and avoid operational delays
- GS6.** identify and troubleshoot common network installation issues using logical problem-solving techniques
- GS7.** make informed decisions under time constraints, especially during on-site troubleshooting
- GS8.** follow safety procedures, industry standards, and best practices to minimize risks in network installation
- GS9.** adapt to new tools, technologies, and evolving industry trends in 5G network deployment
- GS10.** collaborate effectively with cross-functional teams, including network engineers, technicians, and quality inspectors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out power, earthing and RF cabling</i>	15	25	-	10
PC1. ensure the availability of required installation materials as per the installation document, including terminal connectors, thimbles, surge protectors, and cable ties	2	2	-	1
PC2. assess the power requirements of the gNodeB and associated equipment, ensuring compatibility with the available power supply	2	2	-	1
PC3. select appropriate power, earthing, and RF cables based on specifications, load requirements, and environmental conditions	2	2	-	1
PC4. inspect all cables for any physical damage or manufacturing defects and coordinate replacements, if necessary	2	3	-	1
PC5. install and secure power cables from the power source to the equipment, ensuring compliance with safety and regulatory standards	2	3	-	1
PC6. establish proper earthing connections, measuring and maintaining earth resistance values within permissible limits	1	3	-	1
PC7. install RF cables between the gNodeB, other radio equipment, and antennas, ensuring minimal signal loss	1	3	-	1
PC8. route and terminate cables correctly to omnidirectional and sector antennas while allowing flexibility for maintenance and upgrades	1	3	-	1
PC9. conduct continuity and performance tests on installed cables using industry-standard tools and rectify issues if detected	1	2	-	1
PC10. implement shielding and grounding techniques to mitigate electromagnetic interference (EMI) and ensure optimal signal transmission	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install, configure, and commission backhaul connectivity</i>	15	25	-	10
PC11. coordinate with relevant personnel to establish high-throughput Ethernet or fiber-optic backhaul connectivity	2	2	-	1
PC12. deploy fiber optic solutions for backhaul connectivity, ensuring proper termination and splicing techniques	2	2	-	1
PC13. validate and configure backhaul connectivity, ensuring seamless communication between gNodeB and the central office	2	2	-	1
PC14. implement Centralised Unit-Distributed Unit (CU-DU) split architecture for optimized network functionality	2	3	-	1
PC15. configure gNodeB settings to enable integration with the Centralised Unit (CU) and core network	2	3	-	1
PC16. ensure gNodeB visibility in the central unit for remote commissioning, monitoring, and control	1	3	-	1
PC17. install and integrate environmental alarm systems with the central monitoring unit to enable proactive site management	1	3	-	1
PC18. deploy and configure the required Operating System (OS) and Virtual Machine (VM) environments for network operations	1	3	-	1
PC19. verify network synchronization and timing configurations to align with industry standards and optimize performance	1	2	-	1
PC20. conduct functional tests, including throughput and latency checks, to validate end-to-end connectivity and network performance	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6105
NOS Name	Carry out 5G active network installation
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network (Active Components Installation)
NSQF Level	4
Credits	7
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N6246: Follow the Occupational Health and Safety Instructions during Tower Climbing

Description

This unit is about adhering to occupational health and safety guidelines while performing tower climbing tasks to ensure personal safety and compliance with industry standards.

Scope

The scope covers the following :

- Conduct pre-climb tower inspection
- Check safety equipment and assess work site conditions
- Execute tower operations following health and safety protocols

Elements and Performance Criteria

Conduct pre-climb tower inspection

To be competent, the user/individual on the job must be able to:

- PC1.** conduct a visual assessment of the tower using binoculars to check for loose or missing hardware, rust, or structural damage, and ensure necessary repairs are completed before climbing
- PC2.** identify and mitigate climbing obstructions and hazards, such as bird nests, insect infestations, or external attachments
- PC3.** inspect turnbuckles for proper installation and ensure guy wires are tensioned within recommended limits for guyed towers
- PC4.** examine guy preforms, thimbles, and anchor points for signs of corrosion, fraying, or mechanical damage
- PC5.** verify the tower's verticality using a plumb line or inclinometer to detect structural misalignment due to environmental or load stress
- PC6.** report identified tower defects to the relevant authority and confirm corrective actions are implemented before proceeding with climbing tasks

Check safety equipment and assess work site conditions

To be competent, the user/individual on the job must be able to:

- PC7.** inspect ladders, hoisting and rigging equipment, and essential tools to confirm operational readiness and compliance with safety standards
- PC8.** check the condition and availability of Personal Protective Equipment (PPE), including a full-body harness, lanyards, climbing helmets, gloves, and anti-slip boots
- PC9.** verify that a fully stocked first aid kit is accessible at the work site
- PC10.** measure radio frequency (RF) exposure levels using an RF detector to ensure compliance with regulatory limits
- PC11.** identify potential electrical hazards, including proximity to high-voltage power lines, and coordinate necessary precautions

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- PC12.** collaborate with relevant personnel to conduct a Job Hazard Analysis (JHA) and formulate an Emergency Action Plan (EAP)
- PC13.** identify and report unsafe conditions at the work site to the appropriate authority and document corrective actions taken
- PC14.** assess weather conditions using meteorological data and avoid work during thunderstorms, high winds, or adverse environmental conditions
- PC15.** measure wind velocity using an anemometer and ensure work is conducted within permissible wind speed limits
- PC16.** ensure vehicles, machinery, and equipment are positioned at a safe distance from tower structures and live power lines

Execute tower operations following health and safety protocols

To be competent, the user/individual on the job must be able to:

- PC17.** register at the work site and adhere to safety guidelines before commencing tower climbing
- PC18.** secure a full-body harness to designated anchor points and maintain 100% tie-off at all times
- PC19.** utilize a safety cable climb system or double lanyards while ascending, descending, and moving on the tower
- PC20.** use appropriate PPE as per manufacturer guidelines to ensure maximum protection
- PC21.** follow standard tower climbing procedures to prevent falls, slips, or unexpected incidents
- PC22.** maintain continuous communication with the ground crew using a two-way radio
- PC23.** report any health impairments, such as dizziness or fatigue, and cease tower work if under the influence of medication that may affect concentration
- PC24.** work at a safe distance from live power lines or coordinate with authorities to de-energize electrical lines when necessary
- PC25.** place appropriate warning signs near live electrical zones to alert coworkers and prevent accidents
- PC26.** ensure compliance with industry safety regulations and worksite protocols throughout the climbing operation
- PC27.** administer basic first aid for injuries and medical emergencies at the worksite
- PC28.** assist in preparing incident reports by providing accurate details about workplace hazards, injuries, or accidents

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** industry best practices for safe tower climbing and fall protection techniques
- KU2.** the importance of maintaining well-serviced and certified safety equipment before tower operations
- KU3.** essential PPE requirements, including harness types, lanyards, helmets, gloves, and anti-slip boots
- KU4.** procedures for checking and maintaining first aid kits at work sites
- KU5.** effective communication protocols using two-way radios during tower climbing activities
- KU6.** basic first aid techniques applicable to tower climbing incidents
- KU7.** methods for identifying, documenting, and reporting unsafe worksite conditions

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- KU8.** pre-use inspections of PPE, tools, hoisting, and rigging equipment to ensure compliance with safety standards
- KU9.** steps for conducting a Job Hazard Analysis (JHA) and developing an Emergency Action Plan (EAP)
- KU10.** methods for assessing weather conditions and determining when to halt tower climbing operations
- KU11.** risks associated with impaired physical health while working at heights and the impact of medication on concentration
- KU12.** the significance of ongoing safety training and competency enhancement for tower climbers
- KU13.** electrical hazard mitigation strategies and adherence to workplace electrical safety regulations
- KU14.** recommended climbing techniques for different types of telecom towers and related steel structures
- KU15.** industry-specific safety regulations, such as OSHA and local occupational health standards
- KU16.** procedures for preparing and reviewing incident reports to mitigate future risks

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain accurate records of safety inspections, incidents, and maintenance logs
- GS2.** read and interpret safety manuals, regulatory updates, and standard operating procedures
- GS3.** communicate effectively with team members and supervisors regarding safety concerns and work progress
- GS4.** listen attentively to instructions to ensure compliance with safety guidelines
- GS5.** organize and prioritize tasks to complete tower climbing activities safely and efficiently
- GS6.** identify and implement appropriate solutions to safety-related worksite challenges
- GS7.** make quick and informed decisions during emergencies, such as extreme weather or equipment failure
- GS8.** adapt to evolving safety protocols and technology advancements in tower climbing operations
- GS9.** demonstrate situational awareness to recognize potential hazards and prevent accidents
- GS10.** collaborate with coworkers and safety personnel to maintain a risk-free work environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct pre-climb tower inspection</i>	6	10	-	6
PC1. conduct a visual assessment of the tower using binoculars to check for loose or missing hardware, rust, or structural damage, and ensure necessary repairs are completed before climbing	1	2	-	1
PC2. identify and mitigate climbing obstructions and hazards, such as bird nests, insect infestations, or external attachments	1	2	-	1
PC3. inspect turnbuckles for proper installation and ensure guy wires are tensioned within recommended limits for guyed towers	1	2	-	1
PC4. examine guy preforms, thimbles, and anchor points for signs of corrosion, fraying, or mechanical damage	1	2	-	1
PC5. verify the tower's verticality using a plumb line or inclinometer to detect structural misalignment due to environmental or load stress	1	1	-	1
PC6. report identified tower defects to the relevant authority and confirm corrective actions are implemented before proceeding with climbing tasks	1	1	-	1
<i>Check safety equipment and assess work site conditions</i>	12	20	-	5
PC7. inspect ladders, hoisting and rigging equipment, and essential tools to confirm operational readiness and compliance with safety standards	2	3	-	0.5
PC8. check the condition and availability of Personal Protective Equipment (PPE), including a full-body harness, lanyards, climbing helmets, gloves, and anti-slip boots	2	3	-	0.5
PC9. verify that a fully stocked first aid kit is accessible at the work site	1	3	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. measure radio frequency (RF) exposure levels using an RF detector to ensure compliance with regulatory limits	1	2	-	0.5
PC11. identify potential electrical hazards, including proximity to high-voltage power lines, and coordinate necessary precautions	1	2	-	0.5
PC12. collaborate with relevant personnel to conduct a Job Hazard Analysis (JHA) and formulate an Emergency Action Plan (EAP)	1	2	-	0.5
PC13. identify and report unsafe conditions at the work site to the appropriate authority and document corrective actions taken	1	2	-	0.5
PC14. assess weather conditions using meteorological data and avoid work during thunderstorms, high winds, or adverse environmental conditions	1	1	-	0.5
PC15. measure wind velocity using an anemometer and ensure work is conducted within permissible wind speed limits	1	1	-	0.5
PC16. ensure vehicles, machinery, and equipment are positioned at a safe distance from tower structures and live power lines	1	1	-	0.5
<i>Execute tower operations following health and safety protocols</i>	12	20	-	9
PC17. register at the work site and adhere to safety guidelines before commencing tower climbing	1	2	-	0.5
PC18. secure a full-body harness to designated anchor points and maintain 100% tie-off at all times	1	2	-	0.5
PC19. utilize a safety cable climb system or double lanyards while ascending, descending, and moving on the tower	1	3	-	0.5
PC20. use appropriate PPE as per manufacturer guidelines to ensure maximum protection	1	3	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. follow standard tower climbing procedures to prevent falls, slips, or unexpected incidents	1	3	-	0.5
PC22. maintain continuous communication with the ground crew using a two-way radio	1	1	-	0.5
PC23. report any health impairments, such as dizziness or fatigue, and cease tower work if under the influence of medication that may affect concentration	1	1	-	1
PC24. work at a safe distance from live power lines or coordinate with authorities to de-energize electrical lines when necessary	1	1	-	1
PC25. place appropriate warning signs near live electrical zones to alert coworkers and prevent accidents	1	1	-	1
PC26. ensure compliance with industry safety regulations and worksite protocols throughout the climbing operation	1	1	-	1
PC27. administer basic first aid for injuries and medical emergencies at the worksite	1	1	-	1
PC28. assist in preparing incident reports by providing accurate details about workplace hazards, injuries, or accidents	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6246
NOS Name	Follow the Occupational Health and Safety Instructions during Tower Climbing
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network Operation and Maintenance
NSQF Level	4
Credits	3
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N9105: Follow sustainable practices in telecom infrastructure installation

Description

This unit covers sustainable practices in telecom infrastructure installation. It includes identifying components for recycling or refurbishment, ensuring compliance with environmental standards, and promoting sustainability during installation and repair processes.

Scope

The scope covers the following :

- Segregate recyclable and refurbishable components
- Dispose of waste
- Use Energy-Efficient Methods
- Follow environmental standards and compliance guidelines
- Guide team members

Elements and Performance Criteria

Segregate recyclable and refurbishable components

To be competent, the user/individual on the job must be able to:

- PC1.** identify telecom components suitable for recycling or refurbishment
- PC2.** sort electronic and non-electronic waste based on disposal protocols
- PC3.** label and store recyclable and refurbishable components separately

Dispose of waste

To be competent, the user/individual on the job must be able to:

- PC4.** follow approved procedures for the safe disposal of hazardous and non-hazardous waste
- PC5.** coordinate with authorized e-waste recycling units or certified disposal agencies
- PC6.** handle batteries, cables, and electronic circuits safely to prevent environmental hazards

Use Energy-Efficient Methods

To be competent, the user/individual on the job must be able to:

- PC7.** select and use energy-efficient tools and equipment during telecom installations
- PC8.** apply best practices to minimize energy consumption
- PC9.** position and install telecom infrastructure to optimize efficiency and sustainability

Follow environmental standards and compliance guidelines

To be competent, the user/individual on the job must be able to:

- PC10.** adhere to national and international environmental regulations for telecom infrastructure installation
- PC11.** maintain records of waste disposal, recycling, and sustainability measures
- PC12.** assist in periodic audits to assess compliance with green standards
- PC13.** follow workplace sustainability protocols and report non-compliance

Qualification Pack

Guide team members

To be competent, the user/individual on the job must be able to:

PC14. guide team members on sustainable telecom installation guidelines and practices

PC15. encourage environmentally responsible work habits

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the national and international environmental laws and regulations

KU2. e-waste management and recycling policies

KU3. occupational safety and health standards related to environmental practices

KU4. the types of recyclable and refurbishable telecom components

KU5. methods for reducing electronic waste through responsible procurement and reuse

KU6. advancements in eco-friendly telecom infrastructure

KU7. techniques for optimizing energy consumption in telecom operations

KU8. use of renewable energy sources in telecom infrastructure

KU9. best practices for reducing the carbon footprint of telecom installations

KU10. identification and proper disposal methods for hazardous and non-hazardous waste

KU11. the procedures for handling, storing, and transporting waste materials safely

KU12. how to collaborate with authorized agencies for effective waste collection and disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with team members to ensure adherence to sustainable practices

GS2. follow written instructions for waste segregation, disposal, and energy-efficient installations

GS3. use problem-solving skills to address sustainability challenges in telecom installations

GS4. apply numerical skills to track and record environmental compliance data

GS5. organize tasks efficiently to minimize resource wastage and environmental impact

GS6. adapt to new sustainable technologies in telecom infrastructure

GS7. work collaboratively with authorized agencies to ensure proper e-waste disposal

GS8. maintain safety protocols while handling hazardous materials

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Segregate recyclable and refurbishable components</i>	6	12	-	6
PC1. identify telecom components suitable for recycling or refurbishment	2	4	-	2
PC2. sort electronic and non-electronic waste based on disposal protocols	2	4	-	2
PC3. label and store recyclable and refurbishable components separately	2	4	-	2
<i>Dispose of waste</i>	6	10	-	5
PC4. follow approved procedures for the safe disposal of hazardous and non-hazardous waste	2	4	-	2
PC5. coordinate with authorized e-waste recycling units or certified disposal agencies	2	3	-	2
PC6. handle batteries, cables, and electronic circuits safely to prevent environmental hazards	2	3	-	1
<i>Use Energy-Efficient Methods</i>	6	10	-	3
PC7. select and use energy-efficient tools and equipment during telecom installations	2	4	-	1
PC8. apply best practices to minimize energy consumption	2	3	-	1
PC9. position and install telecom infrastructure to optimize efficiency and sustainability	2	3	-	1
<i>Follow environmental standards and compliance guidelines</i>	8	12	-	4
PC10. adhere to national and international environmental regulations for telecom infrastructure installation	2	3	-	1
PC11. maintain records of waste disposal, recycling, and sustainability measures	2	3	-	1
PC12. assist in periodic audits to assess compliance with green standards	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow workplace sustainability protocols and report non-compliance	2	3	-	1
<i>Guide team members</i>	4	6	-	2
PC14. guide team members on sustainable telecom installation guidelines and practices	2	3	-	1
PC15. encourage environmentally responsible work habits	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9105
NOS Name	Follow sustainable practices in telecom infrastructure installation
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N6104.Carry out Rack Level Installation	30	50	0	20	100	25
TEL/N6105.Carry out 5G active network installation	30	50	0	20	100	25
TEL/N6246.Follow the Occupational Health and Safety Instructions during Tower Climbing	30	50	0	20	100	25
TEL/N9105.Follow sustainable practices in telecom infrastructure installation	30	50	0	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	230	0	80	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
TSSC	Telecom Sector Skill Council
TCP	Transmission Control Protocol
IP	Internet Protocol
VLANs	Virtual Local Area Networks
gNodeB	Next Generation Node B
CU-DU	Central Unit - Distributed Unit
RF	Radio Frequency
PTP	Precision Time Protocol
OS	Operating System
VM	Virtual Machine
PPE	Personal Protective Equipment
JHA	Job Hazard Analysis
EAP	Emergency Action Plan

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
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Qualification Pack

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Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (K)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
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Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.